Paper ID: 16281 **Roll No..............................**

END-SEMESTER EXAMINATION, 2023-24
BACHELOR OF TECH. (COMPUTER SC. & ENGG.) (SEMESTER : 04)
CSE 251 : THEORY OF COMPUTATION

Time: 3 Hrs. **Max. Marks: 100**

Note: 1. **All** questions are compulsory.
2. Assume missing data suitably, if any.

CO1: Formulate The Concept Of Automata And Related Terminology.
CO2: Design DFA And NDFA And Conversion From NDFA To DFA.
CO3: Construct Finite Automata Without Output And With Output.
CO4: Implement Regular Expression And Grammar Corresponding To DFA And Vice-versa
CO5: Design Push Down Automata From Context Free Language Or Grammar And Vice-versa .
CO6: Design Turing Machine For Computational Problems, Develop A Clear Understanding Of Un-decidability

CO_S Marks BTL

SECTION-A

All Questions are Compulsory: **(10×4=40 Marks)**

		CO	Marks	BTL
1.	Show regular expressions for (i) Strings of a's and b's with no consecutive b's (ii) Strings of a's and b's containing consecutive b's.	CO1	4	K1
2.	Illustrate Applications and Limitations of Finite Automata.	CO1	4	K2
3.	Define methods for minimization of DFA.	CO2	4	K1
4.	Explain Mealy and Moore machines with suitable examples.	CO2	4	K2
5.	Explain Context Free Grammar (CFG).	CO3	4	K2
6.	Develop a context-free grammar that generate the following languages. i) $\{ w \in \{0, 1\}^* \mid w \text{ contains at least three 1s} \}$ ii) $\{ w \in \{0, 1\}^* \mid w = w^R \text{ and } \lvert w\rvert \text{ is even} \}$	CO3	4	K3
7.	Extend CFG for the regular expression (0+1)*.	CO4	4	K2
8.	Explain the parse tree. Build parse tree for acbd using following grammer: S -> AB A -> c/aA B -> d/bB	CO4	4	K3
9.	Demonstrate Turing machine as a computational machine.	CO5	4	K2
10.	Identify and Explain the purpose of Godel numbering.	CO5	4	K3

SECTION-B

All Questions are Compulsory: **(3×6=18 Marks)**

		CO	Marks	BTL
11.	(a) Examine the language generated by productions: $S \rightarrow aSa \mid aAa$ $A \rightarrow bAb \mid cD$ $D \rightarrow cD \mid c$	CO3	6	K4

--- OR ---

(b) Examine and prove using suitable string that CFG having productions $E \rightarrow E + E \mid E*E \mid E / E \mid id$ is ambiguous.

12. (a) List the steps to convert a CFG into PDA. CO4 6 K4

--- OR ---

(b) Analyse and construct a deterministic pushdown automata for the following languages. Acceptance either by empty stack or by final state.
{a n b an | n ≥ 0}

13. (a) Contrast the Universal TM (Visual Turing machine) and TM. CO5 6 K4

--- OR ---

(b) Simplify the language of NPDA M = ({q0, q1}, {a, b}, {a, z}, δ, q0, z, {q1}), with transitions
(q0; a; z) = {(q0; az)};
(q0; a; a) = {(q0; aa)};
(q0; b; a) = {(q1; Ɛ)};
(q1; b; a) = {(q1; Ɛ)}:
(q1; b; Ɛ) = {(q1; z)}:

SECTION-C

All Questions are Compulsory: (3×10=30 Marks)

14. (a) Let G be the grammar S→ 0B/1A, A → 0/0S/1AA, B → 1/1S/0BB. For the string 00110101, Compare Leftmost Derivation, Rightmost Derivation and Derivation Tree. CO3 10 K4/K5

--- OR ---

(b) Define context free language. Construct context free grammar for language {L= anbmcm+n | m, n >=1}.

15. (a) Discuss that a regular grammar is also a context free grammar, and a context free grammar is also a context sensitive grammar. CO4 10 K4/K5

--- OR ---

(b) Design 2-stack PDA for language L = {anbncn | n ³ 0}.

16. (a) Simplify Church's thesis. CO5 10 K4/K5

--- OR ---

(b) Define Undecidability of Post correspondence problem with suitable examples.

SECTION-D

Attempt the following Question: (1×12=12 Marks)

17. (a) Discuss about modifications to standard Turing Machine. CO6 12 K5/K6

--- OR ---

(b) Discuss Godel Numbering with suitable example.

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